Disruptive Technologies

Agenda Item 3

IAASB Meeting

September 12, 2022

Technology: IAASB Focus

Current and Future Workplan activities

Explore how the IAASB most effectively can respond to technology via new or revised International Standards or non-authoritative guidance

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Investigate disruptive technology trends

Explore technology's effect on audit and assurance – both in the current environment and in the future – in order to be prepared for technology disruption and be able to respond appropriately to support audit and assurance quality

Build	Maintain	Share

Objectives of the Disruptive Technologies initiative

Build	Maintain	Share
Build processes and structures to support the IAASB's disruption initiative	Maintain and improve the IAASB's knowledge about disruption trends and their implications for standard-setting and the public interest	Share knowledge and agenda with stakeholders in the reporting ecosystem to improve audit and assurance quality and strengthen external reporting quality.

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Disruptive Technologies - Activities to date

Build

- Initial research into innovation landscape for Audit and Assurance (Board presentation Jan. 2021)
- Seconded Staff Fellow focused on Disruptive Technology Trends
- Technology Knowledge base built for upskilling staff and Board
- Digital Advisory Group established - first meeting held May 2022

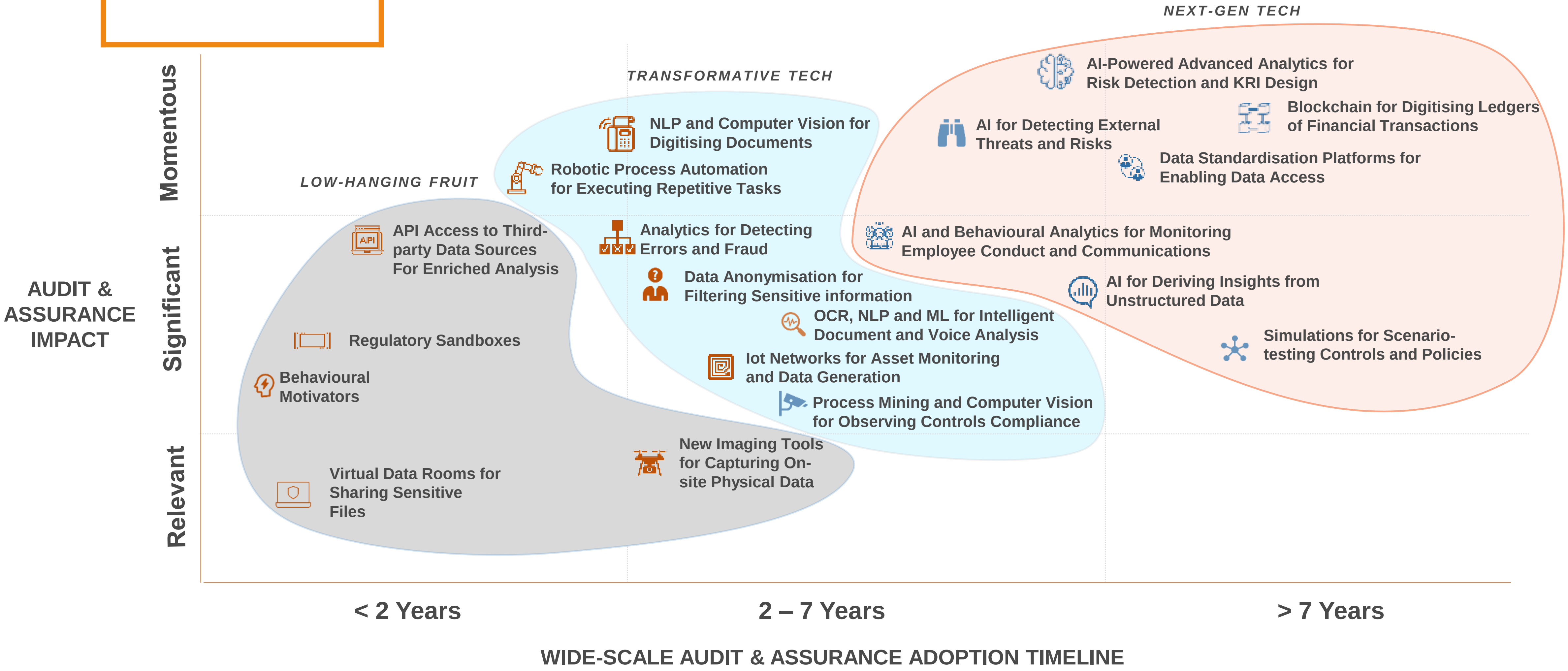
Maintain

- Ongoing outreach and research update activities and interviews
- Board update and awareness sessions
- Interactions with Technology Consultation Group and Current Project teams (e.g., Fraud, Audit Evidence)
- Liaison with IESBA, IAAER and IFAC Technology

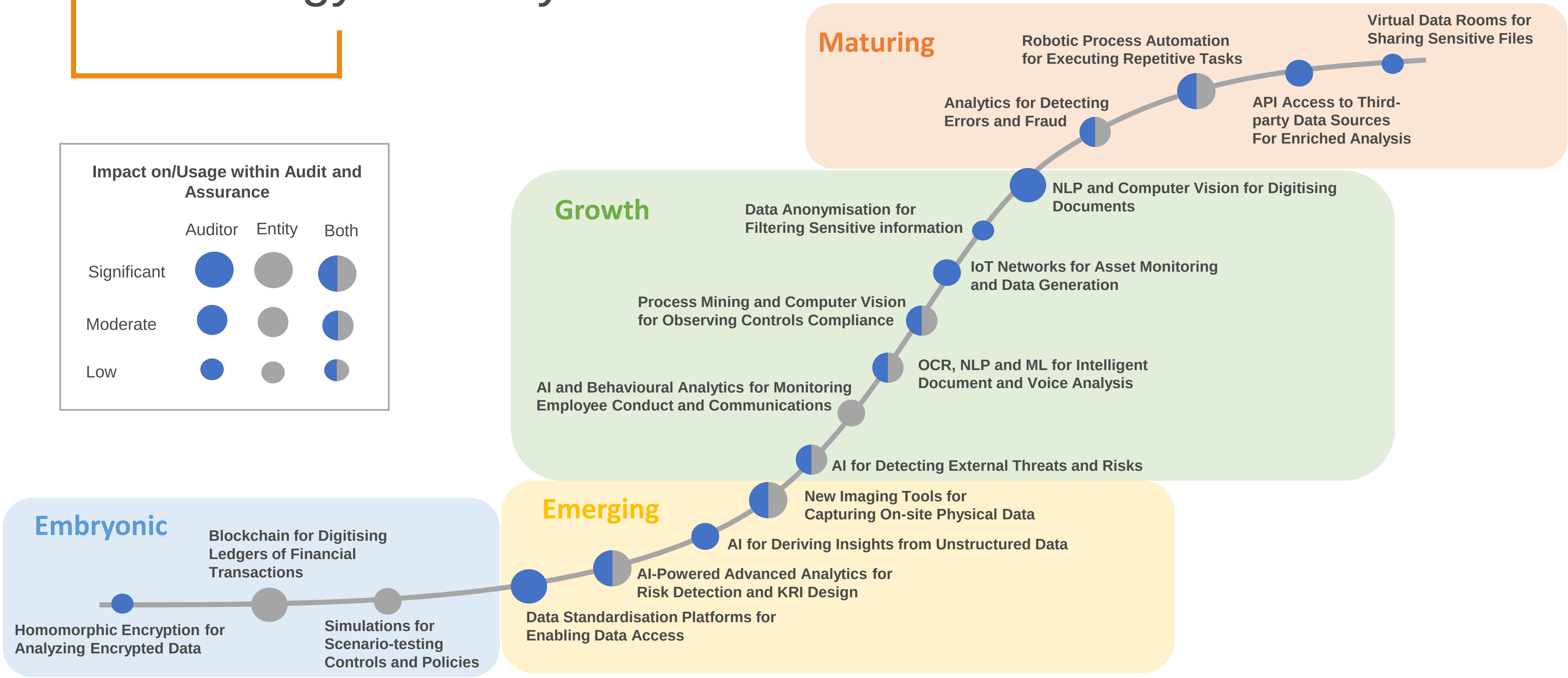
Share

- Disruptive Technologies Roundtables Nov 2020 and Feb 2022
- Regular Digital Technology Market Scan publications
  - Data Standardization
  - Application Programming Interfaces (APIs)
  - Artificial Intelligence : A Primer
  - Natural Language Programming (NLP)
- Thought leadership - Assurance in a Digital Age

2020 Technology Landscape



Technology Maturity Model – 2020 Research



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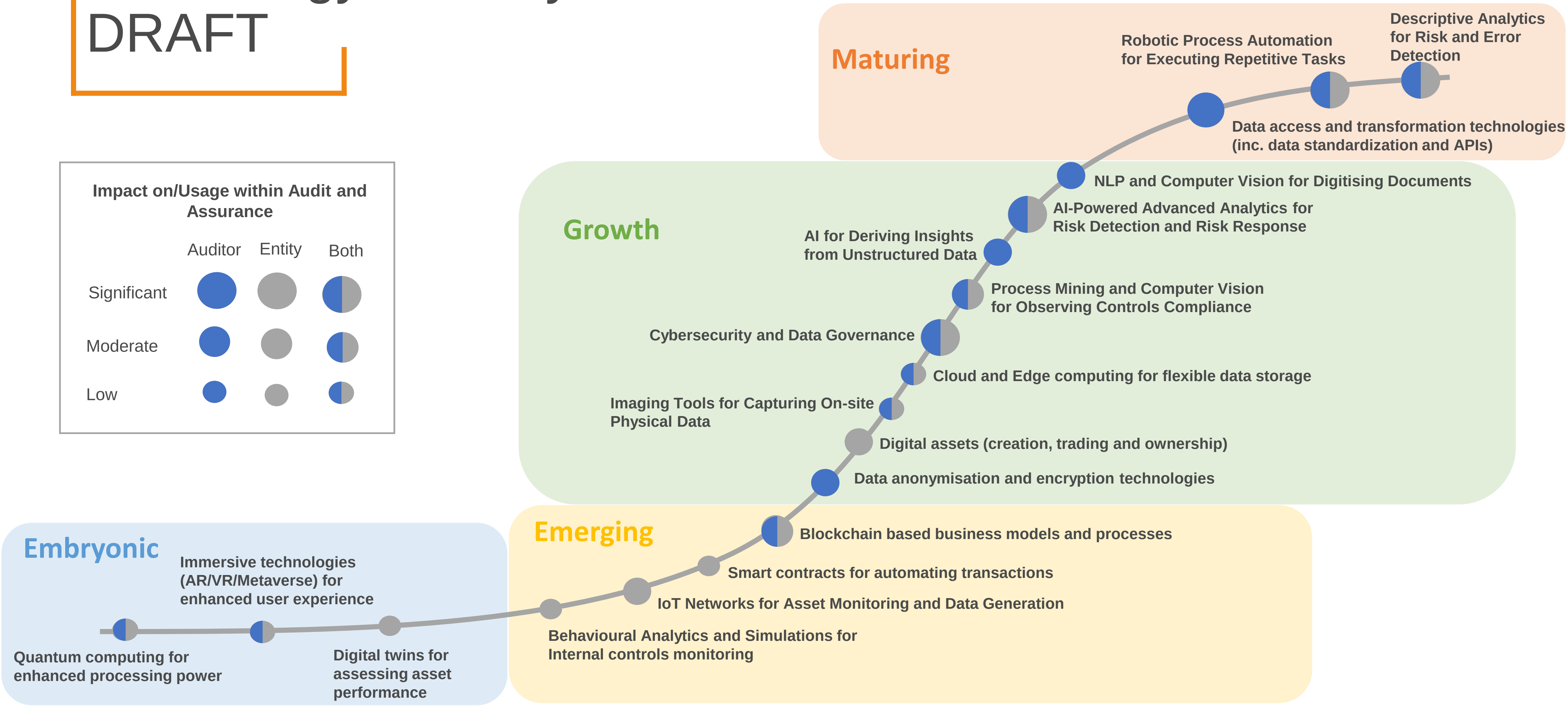
Technology Maturity Model – 2022 Research DRAFT



Trends from research and outreach activities

- Consistent access to appropriate data (including for use in training AI) continues to be a barrier to greater and more widespread use of technology
- Technology adoption lags innovation – people-related factors are a significant contributor including required skills/expertise, confidence, capacity and mindset shift
- Artificial Intelligence and Machine Learning technologies are consistently identified as the most revolutionary and are increasingly used:-
  - To support risk identification and anomaly detection
  - To identify key terms in unstructured documents (e.g., leases, contracts)
  - To perform predictive analytics to support forward looking procedures (e.g., going concern)
- Diverse views exist on the expected impact of Blockchain and related technologies – however Digital Assets are becoming more prevalent
- Remote working continues to influence technology innovations

New technology trends identified

- Cybersecurity and Data Governance
- Cloud and Edge Computing for flexible data storage
- Digital Twins for assessing asset performance
- Quantum Computing for enhanced processing power
- Immersive Technologies (Augmented/Virtual Reality/Metaverse) for enhanced user experience

See Appendix for technology trend terms modified since original research

Matters for Board Discussion

1. The Board is asked for its views on the technology innovation trends highlighted. Of particular interest is which technologies are likely to have the greatest impact on audit and assurance?

Digital Advisory Group

Solon Angel

Janos Barberis

Andrew Miller

Louise Smith

Chris Thatcher

Digital Advisory Group – Items and Themes

Items covered

1) Are the IAASB playing the right role?
2) The future of audit – expert talk
3) Project exploration opportunities

The role of the IAASB	Technology in Auditing Standards	Encouraging the Use of Technology	Convening, Communicating & Collaborating
Other themes	Accessibility and Consumption of Auditing Standards	Barriers to Adoption of Technology - Data	The Attractiveness of the Audit Profession

Breakout Session

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Appendix

Modified technology trend terms

Original Research 2020	Updated Research 2022
• Analytics for Assessing Data and Error and Fraud Detection	• Descriptive analytics on financial and non-financial data
• NLP and Computer Vision for Digitizing Documents • OCR, NLP and ML for Intelligent Document and Voice Analysis	• NLP and Computer vision for Document digitization and enhancement
• AI and Behavioral Analytics for Monitoring Employee Conduct and Communications • Simulations for Scenario Testing Controls and Policies	• Behavioral analysis and Simulations for internal controls monitoring
• Blockchain for Digitizing Ledgers and Financial Transactions	• Blockchain based business models and processes • Digital Assets (creation, trading and ownership) • Smart contracts for automating transactions
• AI-Powered Advanced Analytics for Risk Detection KRI Design • AI for Detecting External Threats and Risks	• AI-Powered Analytics for Risk Identification and Risk Response
• Process Mining and Computer Vision for Observing Controls Compliance	• Process mining for assessing internal controls
• Homomorphic Encryption for Analysing Encrypted Data • Data Anonymisation for Filtering Sensitive Information	• Data anonymization and encryption technologies
• API Access to External Data Sources for Enriched Analysis • Data Standardisation Platforms for Enabling Data Access • Virtual Data Rooms for Sharing Sensitive Files	• Data access and transformation technologies (inc. data standardization and APIs)